

Research Paper

Mixed reality infrastructure based on deep learning medical image segmentation and 3D visualization for bone tumors using DCU-Net

Kun Wang^a, Yong Han^b, Yuguang Ye^{c,d,e}, Yusi Chen^{c,d,e}, Daxin Zhu^{c,d,e}, Yifeng Huang^f, Ying Huang^f, Yijie Chen^g, Jianshe Shi^g, Bijiao Ding^f, Jianlong Huang^{c,d,e,*}

^a Institute of Design, Quanzhou Normal University, Quanzhou 362000, China

^b School of Design, Quanzhou University of Information Engineering, Quanzhou, Fujian 362000, China

^c School of Mathematics and Computer Science, Quanzhou Normal University, Quanzhou, 362001, China

^d Fujian Provincial Key Laboratory of Data-Intensive Computing, Quanzhou Normal University, Quanzhou, 362001, China

^e Key Laboratory of Intelligent Computing and Information Processing (Quanzhou Normal University), Fujian Province University, Quanzhou, 362001, China

^f Department of Diagnostic Radiology, Huaqiao University Affiliated Strait Hospital, Quanzhou, Fujian 362000, China

^g Department of General Surgery, Huaqiao University Affiliated Strait Hospital, Quanzhou, Fujian 362000, China

HIGHLIGHTS

- Automatic segmentation and 3D visualization of CT images by optimizing feature extraction and target space clustering.
- Similarity (DSC), recall (R), precision (P) and three-dimensional vertex distance error (VDE) are combined to evaluate segmentation performance and 3D reconstruction effect.
- The DCU-Net operation model has significant advantages compared with models such as U-Net and Attention-Uet.
- A mixed-reality three-dimensional visualization infrastructure was constructed.
- The facility enhances clinicians' understanding of tumor morphology and spatial relationships, which in turn is expected to facilitate bone tumor clinical practice and improve outcomes.

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ABSTRACT

Objective: Segmenting and reconstructing 3D models of bone tumors from 2D image data is of great significance for assisting disease diagnosis and treatment. However, due to the low distinguishability of tumors and surrounding tissues in images, existing methods lack accuracy and stability. This study proposes a U-Net model based on double dimensionality reduction and channel attention gating mechanism, namely the DCU-Net model for oncological image segmentation. After realizing automatic segmentation and 3D reconstruction of osteosarcoma by optimizing feature extraction and improving target space clustering capabilities, we built a mixed reality (MR) infrastructure and explored the application prospects of the infrastructure combining deep learning-based medical image segmentation and mixed reality in the diagnosis and treatment of bone tumors.

Methods: We conducted experiments using a hospital dataset for bone tumor segmentation, used the optimized DCU-Net and 3D reconstruction technology to generate bone tumor models, and used set similarity (DSC), recall (R), precision (P), and 3D vertex distance error (VDE) to evaluate segmentation performance and 3D reconstruction effects. Then, two surgeons conducted clinical examination experiments on patients using two different methods, viewing 2D images and virtual reality infrastructure, and used the Likert scale (LS) to compare the effectiveness of surgical plans of the two methods.

Results: The DSC, R and P values of the model introduced in this paper all exceed 90%, which has significant advantages compared with methods such as U-Net and Attention-Uet. Furthermore, LS showed that clinicians in the DCU-Net-based MR group had better spatial awareness of tumor preoperative planning.

Conclusion: The deep learning DCU-Net algorithm model can improve the performance of tumor CT image segmentation, and the reconstructed fine model can better reflect the actual situation of individual tumors; the MR system constructed based on this model enhances clinicians' understanding of tumor morphology and spatial

* Corresponding author at: School of Mathematics and Computer Science, Quanzhou Normal University, Quanzhou, 362001, China.

E-mail address: robotics@qztc.edu.cn (J. Huang).

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relationships. The MR system based on deep learning and three-dimensional visualization technology has great potential in the diagnosis and treatment of bone tumors, and is expected to promote clinical practice and improve efficacy.

1. Introduction

Bone tumors mainly refer to tumors that occur in mesenchymal tissues and are rare tumors. Bone tumors often affect the patient's movement and joint function. Therefore, bone tumors need to be diagnosed and treated as early as possible, otherwise they may metastasize to other parts of the body, such as the lungs, other bones, and soft tissues [1]. The treatment of malignant bone tumors is mainly surgical. A study conducted by Cartiaux et al. [2] showed that even experienced surgeons find it difficult to achieve precise tumor resection. Insufficient resection margins can easily lead to tumor recurrence, while excessive resection is traumatic to the patient and is not conducive to postoperative reconstruction and functional recovery of the affected limb. Therefore, surgical resection of bone tumor patients must be combined with the patient's own conditions. Imaging examinations must be fully performed before surgery, and the tumor size and local anatomical structure must be accurately estimated, and the surgical plan can be discussed on this basis.

There are many methods for diagnosing bone tumors. Computed tomography (CT) is one of the main methods for diagnosing bone tumors because of its high resolution and cost-effectiveness. CT examination can display the anatomy of each cross section in layers [3]. However, CT scans can only obtain 2D images. Doctors can only imagine the 3D structure of individual bone tumors based on experience, which can easily lead to surgical accidents due to insufficient preoperative planning. Segmenting bone tumors from CT images is a prerequisite for 3D reconstruction. The study of automatic segmentation has great practical significance and is also an important topic in the field of medical image processing. Traditional image segmentation algorithms such as threshold segmentation and edge detection use a specific attribute of the target to achieve this [4]. The difficulty lies in the fact that the pixel grayscale distinguishability between bone tumors and surrounding tissues is low and there is no generally clear boundary, so these methods are not suitable for automatic segmentation of bone tumor CT images.

In recent years, imaging and digital technology have continued to advance, making it possible to accurately calculate the gross tumor volume (GTV) before surgery [5]. Traditionally, tumor surgeons view two-dimensional (2D) medical images on computer workstations. Image processing software can create 3D models from 2D imaging data sets. The premise for creating 3D models is the accurate segmentation of medical images. The goal of medical image analysis is to provide radiologists and clinicians with efficient diagnosis and treatment processes [6]. Since medical images contain a large amount of data and are subject to noise interference, it is very important to process the images and extract effective information from them [7]. Image segmentation methods based on deep learning can classify and predict images pixel by pixel by training convolutional neural networks (CNNs), providing a new idea for complex medical image segmentation and showing great advantages [8–10]. U-Net is a very outstanding CNN algorithm model in the field of medical image segmentation. It is a semantic segmentation model using convolutional layers. The model structure was first proposed by Ronnerberger et al. [11] in 2015. Currently, U-Net and U-Net-like models have been successfully used to segment various biomedical images, such as liver [12], brain tumors [13], blood vessels [14], and oral cavity [15]. The network structure of U-Net is symmetrical, and the decoder is also composed of four blocks. Finally, a convolution operation generates the final segmentation. However, in medical images, the objects of interest sometimes have different scales. Therefore, in order to achieve better segmentation accuracy, the network structure needs to have the ability to analyze images of different scales. Based on the

advantages and challenges of U-Net, Szegedy et al. [16] introduced an innovative architecture based on the U-Net architecture, namely the MultiResUnet architecture. The application of this inception-like module enables the U-Net architecture to connect features learned from images of different scales while reducing computational and memory requirements. In addition to the MultiRes module, they also made some improvements to the jump connection (Res-Path) between the encoder and decoder. In addition, Schlemper et al. [17] used an attention gate module embedded in U-Net to eliminate the necessity of positioning, but this mechanism failed to focus on the clustering characteristics of bone tumors in the image space. By comparing U-Net and MultiResUnet, Lou et al. [18] pointed out that MultiResUnet obtained better output than the U-Net model, but in some cases, MultiResUnet may not work well in feature extraction. To this end, they creatively developed a new architecture. They used a dual-channel block, whose structure is basically the same as that used in U-Net and MultiResUnet, but achieved more accurate feature extraction. Based on the research of Lou et al., this study proposed a dual dimensionality reduction and channel attention gate U-shaped network (DCU-Net), which can better realize automatic segmentation and 3D reconstruction of bone tumors by optimizing feature extraction and improving target space clustering ability.

In addition, although the precise segmentation and 3D reconstruction we completed based on DCU-Net help clinicians understand the tumor morphology and spatial relationship, the 3D model displayed on the computer 2D flat screen before the actual operation still lacks parallax and depth perception [19]. The challenge is that when doctors observe patients from multiple angles, they need to maintain a mental model that corresponds to the actual observation and accurately remember its spatial direction. This requires doctors to have a high degree of spatial imagination and memory, and to be able to quickly identify and understand the patient's anatomical structure from different perspectives. Therefore, after completing precise segmentation and 3D reconstruction based on DCU-Net, we further combined the immersive technology of mixed reality (MR). The reconstructed virtual 3D bone tumor model of the patient is mapped to its anatomical structure. Further explore the bone tumor diagnosis and treatment infrastructure based on DCU-Net and MR. In summary, this study aims to: (1) Explore the use of DCU-Net model to accurately identify and segment tumor boundaries, and then use 3D reconstruction technology to generate a physical 3D bone tumor model to improve diagnostic accuracy. (2) Based on DCU-Net and the 3D bone tumor model, a 3D holographic MR is constructed to increase the doctor's spatial awareness of the tumor location and direction, help doctors better understand the spatial relationship between bone tumors and surrounding tissues, and improve the effectiveness of surgical planning.

2. Methodology

2.1. Model Selection

U-Net is an excellent and popular architecture in the field of medical image segmentation. MultiResUnet based on U-Net can provide better output than U-Net because it can provide features of different scales [18]. However, when dealing with some difficult medical image cases, MultiResUnet may not achieve the expected performance. The MultiRes module of this network was originally designed to capture features of different scales in order to better distinguish the target object from the image. In order to improve this performance, we improved the MultiRes module to obtain more accurate and effective features. Based on the principle of deep learning, this study proposed a new U-Net model,

namely the DCU-Net model. This model introduces dual dimensionality reduction technology and channel attention gating mechanism to enhance the network's ability to capture key features in bone tumor images. With this improvement, the DCU-Net model can more effectively handle challenging image cases and improve the accuracy and reliability of segmentation.

The DCU-Net proposed in this study consists of an encoding path and a decoding path. The encoding path consists of a convolutional unit and a dual dimensionality reduction module. The convolutional unit is a convolutional layer, batch normalization (BN), and a ReLU activation function to form a (Conv-BN-ReLU) $\times 2$ structure. The output of each convolutional unit is subjected to feature dimensionality reduction by the dual dimensionality reduction module [2]. The decoding path consists of upsampling, convolutional units, and channel fusion to reconstruct feature maps. A channel attention gating module is added to the encoder-decoder connection to filter redundant features. Finally, a convolutional layer with a kernel of 1×1 is passed, and a pixel classification probability map is output with a Sigmoid function. The overall structure of bone tumor image segmentation based on DCU-Net is shown in Fig. 1.

2.2. Double dimensionality reduction module

In the encoding path of a deep learning network, shallow features obtained by step-by-step dimensionality reduction are crucial for the final pixel-level recognition. These features provide a basic understanding of the objects in the image and are the key to accurate segmentation. In order to capture finer details of bone tumors in the shallow feature sampling stage, this study introduced a dual dimensionality reduction module to achieve deeper feature dimensionality reduction. The design of this module allows the network to feed the feature map of the previous layer into the maximum pooling layer and the convolution layer at the same time. Both layers use a 2×2 convolution kernel with a step size of 2 to ensure that the resolution of the output feature map remains consistent. In this way, the maximum pooling layer can capture the global information of the image, while the convolution layer can retain more local details. Finally, the outputs of the two layers are merged by element-by-element addition, and the merged result is then injected into each stage of the encoding path. The design of this dual dimensionality reduction module not only enhances the network's ability to capture detailed features, but also increases the expressiveness

of features by merging outputs from different levels. This structural innovation helps to improve the network's analysis and segmentation accuracy of complex medical images. The module structure diagram shown in Fig. 2 illustrates this process.

2.3. Channel attention gating module

This study proposes a channel attention gating module that automatically screens shallow features and embeds it in the encoder-decoder connection to improve the network's ability to focus on learning segmentation targets. The attention coefficient $\alpha \in [0, 1]$ in the module is used to suppress the feature channel response in the shallow feature X_d that is weakly correlated with the segmentation target. The final output of the module is the multiplication of α and X_d . The calculation method of the attention coefficient α is shown in Equations (1)-(3).

$$Q = \omega_u^T X_u + \omega_d^T X_d + b_q \quad (1)$$

$$g_i = \frac{1}{H \times W} \sum_{j=1}^H \sum_{k=1}^W Q_i(j, k) \quad (2)$$

$$\alpha = \sigma_2(\omega_p \sigma_1(\omega_p + b_p) + b_p) \quad (3)$$

In the formula, Q linearly maps the two sets of input feature maps to a feature space with a resolution of $H \times W \times C_{int}$, where C_{int} is defined as one quarter of C_d . g is the initial channel feature vector obtained after Q global average pooling (GAP). σ_1 is the ReLU activation function. σ_2 is the Hard-Sigmoid function, which has high computational efficiency and can balance the additional computational overhead generated by the two fully connected layers as presented by Equation (4). The overall structure of the module is shown in Fig. 3.

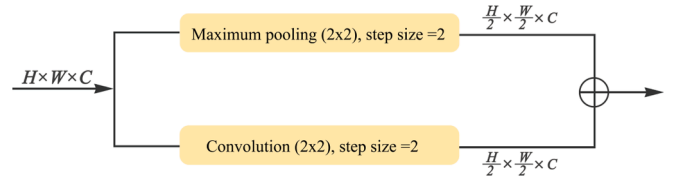


Fig. 2. The structure of dual dimensionality reduction module.

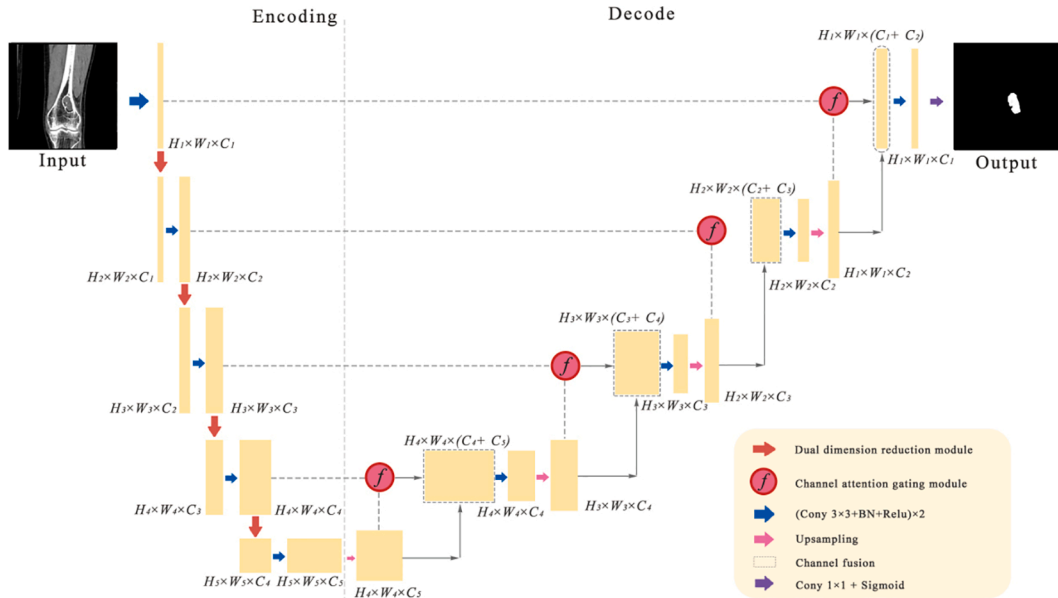


Fig. 1. The overall structure of encoding and decoding paths in bone tumor image segmentation process based on DCU-Net.

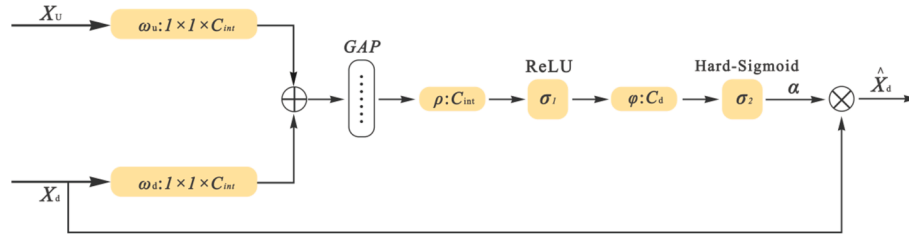


Fig. 3. The structure of channel attention gate module. (X_d is the convolution unit output of the encoding path, the size is $H \times W \times C_d$; X_u is the characteristic diagram of the corresponding decoding layer, the size is $H \times W \times C_u$).

$$\sigma_2(x) = \max\left(0, \min\left(1, \frac{x+1}{2}\right)\right) \quad (4)$$

3. Experiment

3.1. Data collection and experimental settings

The dataset of this study mainly comes from Huaqiao University Affiliated Strait Hospital in China, and some images in the dataset are shown in Fig. 4. We selected 50 cases of bone tumor CT data stored (labeled as CT-50). To ensure the accuracy and availability of the data, we performed detailed layer-by-layer manual annotation of these bone tumor regions. Each CT sample contains 163 to 482 images with a uniform spatial resolution of 512 pixels $\times$ 512 pixels.

3.2. Data preprocessing

The CT-50 dataset was converted into two-dimensional images, and bone tumor slices were selected from them. The dataset was divided into 40 cases for training and 10 cases for testing. The preprocessing steps include: first, the image was scaled to 256 $\times$ 256 pixels using the center cropping technique to preliminarily exclude areas irrelevant to the segmentation target; second, the histogram equalization technique was applied to improve the image contrast and maximize the information entropy; finally, the pixel values were normalized.

3.3. Network training

The Adam optimizer was used for training optimization, with a batch size of 40 and 200 iterations. Data enhancement was added during the

training process to improve the generalization ability of the network. The image enhancement strategies included: random rotation, random scaling, random translation, random horizontal mirroring, and random vertical mirroring. Since bone tumors are small targets in images, the Dice loss function was used in the training process to solve the problem of pixel class imbalance [20]. The loss function is defined in Equation (5).

$$L_{Dice} = 1 - \frac{2 \times \sum (p \times T) + \epsilon}{\sum p + \sum T + \epsilon} \quad (5)$$

In the formula, p is the probability map output by the network, $p_i \in [0, 1]$. T is the binary image of the real label of the bone tumor area, $T_i \in \{0, 1\}$. ϵ is the minimum constant.

3.4. Performance evaluation indicators

The proposed method is compared with three networks formed by U-Net [11], Attention-Unet [17] and the improved method combined with Res-Net [21]. In addition to the commonly used evaluation indicators of dice similarity coefficient (DSC), recall (R) and precision (P) in medical image segmentation, we also added three-dimensional vertex distance error (VDE) to evaluate the 3D reconstruction model [22]. Among the multiple indicators for evaluating the segmentation effect, a higher DSC score means a higher consistency between the segmentation result and the true annotation. An increase in the R score indicates that the network is more sensitive to the recognition of bone tumor features. An increase in the P score reflects that the network is more accurate in identifying bone tumor pixels. A decrease in the VDE score indicates that the 3D reconstruction model is more accurate. To calculate the value of VDE, it is necessary to: (1) reconstruct the segmentation result and the binary

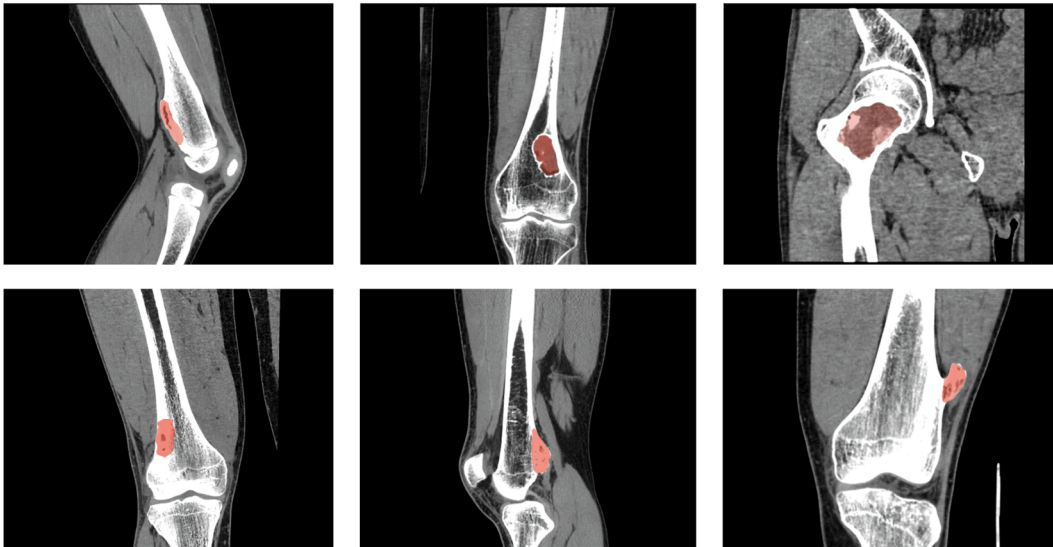


Fig. 4. Display of some images from a CT dataset of 50 cases of bone tumors.

image sequence of the true label into a 3D model using the marching cube (MC) algorithm; (2) align the two 3D models using the iterative closest point algorithm and generate the same number of vertex pairing groups through 3D resampling; (3) calculate the Euclidean space distance between each sampled vertex group, and finally obtain the average distance as the VDE value [26].

3.5. Experimental results

As shown in Table 1, the segmentation results of 10 test samples after the four networks were trained with all training set samples (40 samples). The average values of DSC, R and P3 indicators of this method reached 92.23%, 91.16% and 93.84% respectively, and the standard deviation was the smallest among all experimental networks. The experimental results of this method are better than those of the other three comparison networks.

Table 2 compares the segmentation effects of the four networks on 10 test samples when the number of training samples is reduced to 24. The results show that although the performance of all networks has declined, the method in this study outperforms other networks in terms of average DSC, R and P. At the same time, the standard deviations of the three indicators DSC, R and P are also reduced.

In this study, it is worth noting that in addition to the dice similarity coefficient (DSC), recall (R) and precision (P) commonly used in medical image segmentation, we also The three-dimensional vertex distance error (VDE) is added to evaluate the 3D reconstruction model. Among the multiple indicators to evaluate the segmentation effect, a higher DSC score means a higher consistency between the segmentation results and the real annotations. In contrast to the DSC score, a decrease in the VDE score indicates a higher accuracy of the 3D reconstructed model. Table 3 shows the 3D reconstruction model errors obtained by the four networks. The mean VDE of this method is reduced to 1.19, with the smallest standard deviation, which is better than the other three network methods. In order to intuitively demonstrate the effect of the 3D reconstruction model, we reconstructed the 3D model based on the above method and performed visual operations. Fig. 5 shows the visual comparison of VDE indicators using four different networks. It can be seen more intuitively from the sample diagram that the DCU-Net explored in this study significantly improves the accuracy of 3D reconstruction of bone tumors.

3.6. Evaluation of the effectiveness of mixed reality technology

Although the precise segmentation and 3D reconstruction we completed based on DCU-Net help clinicians understand the tumor morphology and spatial relationship, before the actual operation, the 3D model displayed on the computer 2D flat screen still lacks depth perception and parallax. When observing the patient from different angles, the surgeon must mentally retain the spatial direction of the matching 3D model. Therefore, after completing the precise segmentation and 3D reconstruction based on DCU-Net, we further combined the immersive technology of mixed reality (MR). We imported the patient's reconstructed 3D bone tumor model into the MR system and then mapped it to its anatomical structure to further explore the bone tumor diagnosis and treatment infrastructure based on DCU-Net and MR.

Evaluation methods: Two surgeons with more than 3 years of clinical experience performed clinical examinations and simulated preoperative

Table 2

Segmentation results of 10 test samples after 24 training samples.

Network	DSC / % ($\bar{x} \pm s$)	R / % ($\bar{x} \pm s$)	P / % ($\bar{x} \pm s$)
U-Net	67.26 $\pm$ 17.47	66.93 $\pm$ 19.87	66.93 $\pm$ 19.87
Attention-Unet	72.65 $\pm$ 12.22	71.45 $\pm$ 18.41	71.45 $\pm$ 18.41
Res-Net	59.91 $\pm$ 17.20	53.38 $\pm$ 19.20	53.38 $\pm$ 19.20
Our method	84.33 $\pm$ 12.02	84.79 $\pm$ 16.88	84.79 $\pm$ 16.88

Table 3

Result depicting 3D reconstruction model error.

Network	VDE
U-Net	3.19 $\pm$ 2.83
Attention-Unet	2.86 $\pm$ 2.47
Res-Net	2.60 $\pm$ 2.04
Our method	1.19 $\pm$ 0.08

planning and assessment for each patient using two different methods: viewing with Microsoft HoloLens 2 and viewing with 2D images. For each method, the surgeon completed a qualitative survey: a Likert scale (LS) questionnaire to assess his perception of spatial awareness of bone tumors and effectiveness of surgical planning. We modified the LS questionnaire used in the study by Wong [19]. LS has a five-point scale, and satisfaction is represented by 1 to 5 points from low to high. This study used a Likert scale questionnaire to evaluate and compare the doctors' understanding of bone tumor space when using the two methods, thereby reflecting the effectiveness of the two surgical planning methods. The LS questionnaire uses a five-point rating scale to comprehensively evaluate physicians using the following specific indicators: spatial visualization, location and identification, surgical incision, spatial orientation, depth perception, and Effectiveness of tools. It should be noted that spatial visualization simply refers to the ability to observe a two-dimensional figure and imagine what it would look like when it is unfolded in three-dimensional space, as well as the ability to rotate and manipulate objects in the mind without physical objects. Spatial orientation refers to the ability to identify the position or direction of an object in space in the mind, whereby these abilities are crucial for clinicians.

We also conducted an ablation experiment. As shown in Table 4, in terms of the mean of the ablation experiment results of this method, the key variables in the method are the dual dimensionality reduction module (DDR) and the channel attention gating module (GAG). It can be seen from the results that when DDR and GAG are used at the same time, the DSC and P values reach the highest scores, while the VDE value drops to the lowest.

3.6.1. Preparation of MR system application

We built a mixed reality (MR) infrastructure and developed corresponding MR holographic applications for each case. In addition to the 3D model mainly based on VDE reconstruction, we also integrated CT and MRI image data. From these multi-source images, we created a 3D bone model and further generated a CAD model from the 2D medical images. Combined with the VDE images, we used computer graphics software to produce a rendered model with color, material, texture, and lighting effects to simulate the real reflection of light on the surgical object. Subsequently, we developed a user interface using a 3D engine, which is widely used in virtual simulation and integrates 2D medical images, digital medical information, and 3D models. The holographic application calculated the visual presentation of the scene and generated interactive 3D holographic images. Finally, these contents were packaged into a digital package and loaded into the Microsoft MR-Hololens 2 device.

Two attending physicians performed clinical evaluations on two patients with bone tumors and developed surgical plans. The orthopedics doctors used two methods, 2D and MR 3D holograms, respectively.

Table 1

Segmentation results of 10 test samples after 40 training samples.

Network	DSC / % ($\bar{x} \pm s$)	R / % ($\bar{x} \pm s$)	P / % ($\bar{x} \pm s$)
U-Net	76.41 $\pm$ 10.26	79.07 $\pm$ 14.62	77.28 $\pm$ 9.41
Attention-Unet	81.22 $\pm$ 5.94	80.82 $\pm$ 9.97	79.79 $\pm$ 6.50
Res-Net	80.13 $\pm$ 6.34	81.02 $\pm$ 11.84	81.27 $\pm$ 6.89
Our method	92.23 $\pm$ 4.97	91.16 $\pm$ 7.67	93.84 $\pm$ 4.12

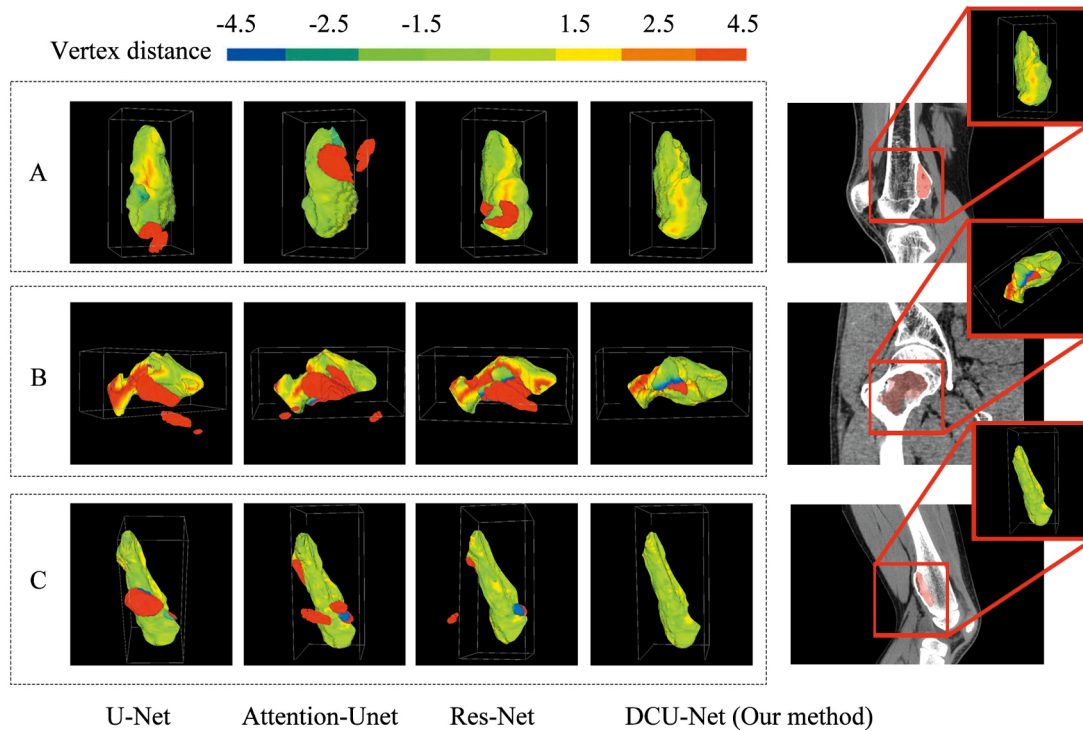


Fig. 5. Visual comparison of 3D reconstructions of bone tumors (VDE index). (The blue direction indicates a concave error, and the red direction indicates a convex error). (5A) Samples with relatively accurate subjects; (5B) Samples with relatively large VDE values; (5C) Samples with relatively small VDE values.

Table 4
Ablation experiment results.

DDR	CAG	DSC/%	R / %	P / %	VDE
×	×	76.41	79.07	77.28	3.19
✓	×	81.22	80.82	80.90	3.77
×	✓	80.13	89.46	74.16	2.74
✓	✓	92.23	91.16	93.84	1.19

In the 2D method, doctors rely on 2D image data, build a 3D model in their mind, and match it to the patient's body (Fig. 6A). In contrast, in the MR 3D hologram method (Fig. 6B), doctors are able to use the spatial observation mode to fully examine the patient's tumor area before surgery and observe the 3D hologram directly on the patient through a head-mounted display device. Microsoft HoloLens 2 is equipped with a camera that tracks the doctor's hand movements. In the interactive display mode, doctors can interact with the hologram in real time through gestures, or manually align the hologram with the patient's actual anatomical structure.

3.6.2. MR application test results

This study recorded and compared the results of the LS questionnaire for preoperative surgical planning of patients with bone tumors using screen 2D and MR 3D hologram methods, and we summarized them as median and interquartile range (IQR). Qualitative investigation based on the LS questionnaire showed (Table 5): Compared with the screen 2D group, the MR 3D group achieved better results in all six indicators of tumor spatial awareness. Therefore, we believe that the effect of the MR system on diagnosis and treatment is more obvious.

4. Discussion

Efficient and accurate segmentation and reconstruction of three-dimensional models of bone tumors from two-dimensional images plays a key role in accurately assessing individual anatomical structures,

guiding the formulation of surgical plans, and designing innovative implant devices [27] or manufacturing mechanical implant structures [23–25]. However, the current method of manually segmenting medical images layer by layer, which is commonly used in the industry, not only requires the operator to have computer operation skills and advanced medical knowledge, but also is complex and time-consuming, and often fails to meet the urgent clinical needs. Due to the variable shapes of bone tumors in two-dimensional images and the low distinguishability between tumors and surrounding tissues, the U-Net-based deep learning segmentation method is not accurate and stable enough, resulting in large errors in the 3D reconstruction model. Based on the U-Net basic architecture and a hospital dataset, this study proposes an improved network (DCU-Net) that combines dual dimensionality reduction and channel attention gating, which helps to improve the accuracy of bone tumor image segmentation and 3D reconstruction [35,36].

The data in Tables 1 and 2 show that DCU-Net can still obtain higher DSC, R, and P scores than other methods even when the training samples are reduced. This shows that DCU-Net can enhance the bone tumor segmentation effect of U-Net under training data of different scales and show higher robustness. The data in Table 3 points out that DCU-Net has a smaller average error in 3D model reconstruction and is closer to the real labeled 3D model. In contrast, the VDE values of the bone tumor 3D models obtained by the other three comparison networks are higher, which may mean that there are more omissions in the segmentation process. This difference may originate from the max pooling dimensionality reduction method used by the contrasting network, which only represents the maximum value in a local area and may lose key detailed features when compressing the feature space. The dual dimensionality reduction rule adopted by DCU-Net combines maximum pooling and convolution operations, which not only captures local features but also takes into account global information. This method enables the network to learn local and global features simultaneously during the dimensionality reduction process, thereby improving the ability to extract detailed features of bone tumors and reducing omissions in the segmentation process. The experimental results further confirmed that only

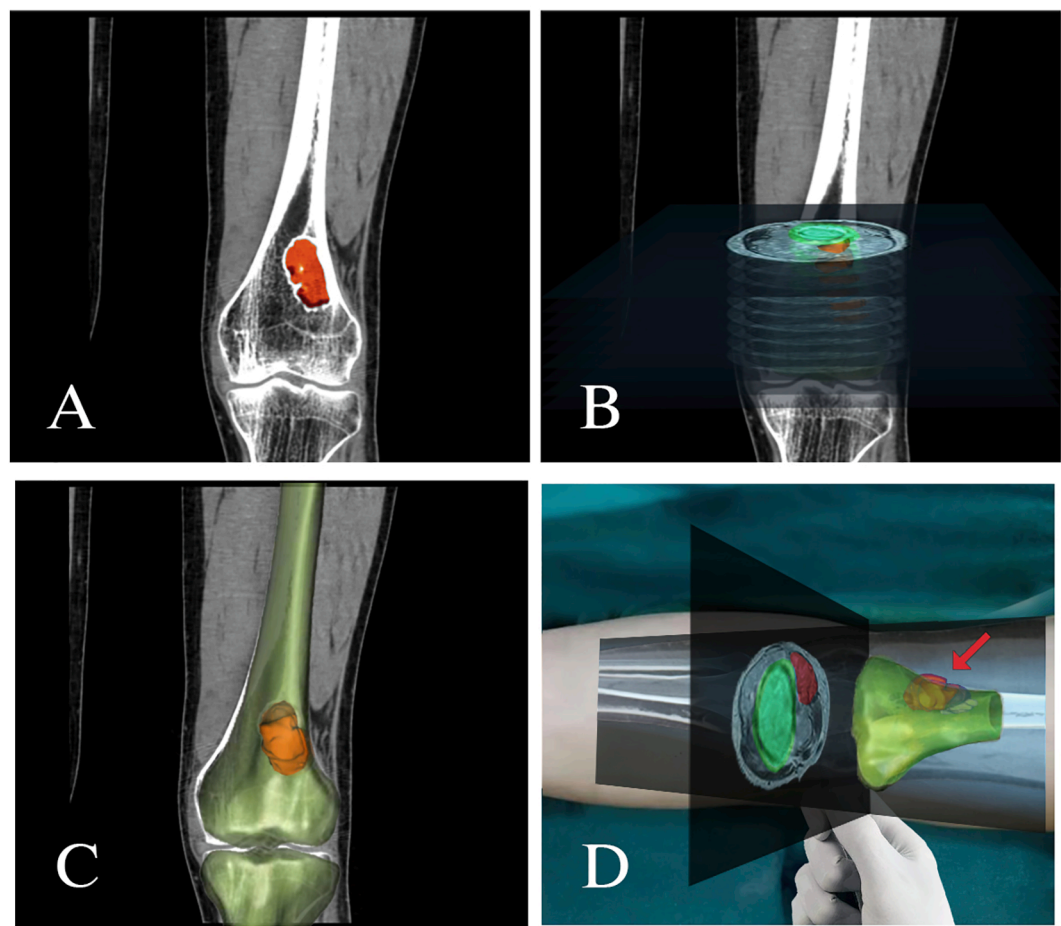


Fig. 6. Display effect of mixed reality technology (A: 2D image of CT; B: 3D stereoscopic effect formed by slicing different color regions; C: 3D models; D: MR 3D display).

Table 5
Results of Likert Scale Questionnaire Scores.

Standards	2D Group, Median (IQR)	MR 3D Group, Median (IQR)
Spatial visualization	2 (2)	5 (5)
Location and identification	2 (2)	4 (4)
Surgical incision	2 (2)	5 (5)
Spatial orientation	2 (2)	5 (5)
Depth perception	1 (1)	4 (4)
Effectiveness of tools	2 (2)	4 (4)

using DDR to improve U-Net significantly improved the DSC mean, R, P and VDE. This indicates that double dimensionality reduction may not only enhance the extraction of detailed features of bone tumors, but also introduce redundant information in the feature space. When CAG is applied alone, the network’s oversensitivity to bone tumor features leads to incorrect identification of background pixels as bone tumors, increasing the misjudgment rate. However, when DDR and CAG are used together (i.e., the improved model), the DSC, R, and P values are significantly improved, and the VDE value is also significantly reduced. This shows that the method proposed in this study that combines dual dimensionality reduction and channel attention gating can only achieve optimal performance when applied simultaneously.

In addition, in complex bone tumor surgeries, surgical planning is a major challenge [29]. Before surgery, surgeons need to filter 2D images in a mental model and superimpose a 3D model on the patient’s body part to be operated on before deciding the surgical entry path and

cutting technique of the surrounding bones [30]. Therefore, after using the above method to enhance the extraction of bone tumor detail features and reduce segmentation omissions, this study further applied the 3D extracted by this method to MR and conducted experiments. This method does not require surgical exposure. Preliminary results show (Table 4): Compared with conventional two-dimensional observation methods, three-dimensional holograms in mixed reality (MR) technology have obvious advantages in improving the three-dimensional visualization [32] and spatial perception of patients’ bone tumors.

In summary, the mixed reality infrastructure for three-dimensional visualization [33] of bone tumors based on deep learning [28] helps to increase doctors’ spatial awareness of tumor location and direction, helps doctors better understand the spatial relationship between bone tumors and surrounding tissues, and improves the effectiveness of surgical planning. However, this study also faces some limitations: when performing DCU-Net image segmentation, manual intervention is currently required to identify and select image layers containing bone tumors. The fully automatic positioning of bone tumors in CT volumes is yet to be achieved. Furthermore, although MR technology has shown positive signs in assisting orthopedic tumor surgical planning, this study is only a preliminary exploration. Only a few surgeons conducted preliminary evaluations on limited cases in the experiment, which limited the generalizability and reliability of the results. Therefore, more surgeons and patients are needed to draw statistical significance for the better results that mixed reality technology may bring. Finally, the feasibility of MR reality infrastructure combined with a cybernetical intelligence platform [31] and 3D visualization [34] in actual clinical applications, user acceptance, and integration with other surgical tools

still need further research and verification.

5. Conclusion

In order to explore the mixed reality technology of bone tumor image segmentation and 3D visualization diagnosis and treatment based on deep learning, this study proposed a U-Net network combining double dimensionality reduction and channel attention gating, namely DCU-Net. Double dimensionality reduction can enhance the effective information extraction in the shallow feature space, and channel attention gating can better filter redundant features and maintain the spatial integrity of bone tumors. Experimental results show that the improved model achieves higher accuracy of bone tumor image segmentation and 3D reconstruction. In addition, we also use the reconstructed 3D features of bone tumors and load them into the Microsoft MR-Hololens 2 device, which preliminarily confirms that this technology helps to improve clinicians' spatial awareness and to some extent makes up for the limitations of two-dimensional image feedback, thereby helping to improve the effectiveness of surgical planning in orthopedic tumor surgery. This study is the first study to combine deep learning medical image segmentation and virtual reality technology to promote the diagnosis and treatment of orthopedic tumors by surgeons. This technology not only helps pathologists improve image segmentation accuracy, but also helps clinicians gain better spatial awareness before surgery. The potential benefits of deep learning and 3D visualization infrastructure for bone tumors are that it is likely to improve the accuracy, safety and efficiency of surgery.

Ethics approval

All human subjects in this study have given their written consent for the participation of our research.

Data availability statement

Bone tumor image data was used to support the results of this study, conducted by Huaqiao University Affiliated Strait Hospital, Provided by Quanzhou, Fujian, China, under license and so cannot be made freely available.

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CRediT authorship contribution statement

Kun Wang: Writing – review & editing, Writing – original draft. **Yong Han:** Writing – original draft. **Yuguang Ye:** Resources, Project administration. **Yusi Chen:** Formal analysis. **Daxin Zhu:** Formal analysis, Data curation, Conceptualization. **Yifeng Huang:** Investigation. **Ying Huang:** Investigation. **Yijie Chen:** Formal analysis, Data curation, Conceptualization. **Jianshe Shi:** Formal analysis, Data curation, Conceptualization. **Bijiao Ding:** Investigation, Data curation. **Jianlong Huang:** Visualization, Validation, Supervision, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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